Foundation of Blockchain Technology (IS501)
Pre-Reading Material – Lecture 30

Encryption Schemes-II

Estimated Reading Time: 15–20 minutes

Introduction

Blockchain applications may need to protect sensitive information while maintaining the transparency and verifiability of blockchain records. Different encryption schemes and privacy-enhancing techniques can help achieve these goals.

Common Encryption Algorithms

A. Advanced Encryption Standard (AES)

AES is a widely used symmetric encryption algorithm. It uses the same secret key for encryption and decryption.

AES is suitable for protecting large amounts of information, such as medical records, confidential documents and files associated with blockchain applications.

B. RSA Encryption

RSA is a public-key cryptographic algorithm that can be used for encryption as well as digital signatures through distinct procedures.

In an encryption scenario, the sender uses the recipient's public key to encrypt information, and the recipient uses the corresponding private key to decrypt it.

RSA encryption is generally less efficient than symmetric encryption for large data volumes. In practice, hybrid encryption is often preferred.

C. Elliptic Curve Cryptography (ECC)

Elliptic curve cryptography is based on mathematical structures involving elliptic curves over finite fields. It supports several cryptographic operations, including key agreement and digital signatures.

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ECC is widely used in blockchain systems for digital signatures and can also support key establishment mechanisms used with encryption systems.

ECC itself is a family of cryptographic techniques rather than a single encryption algorithm.

D. Hybrid Encryption

Hybrid encryption combines asymmetric and symmetric cryptography.

A typical process involves:

1. Generating a random symmetric key.
2. Encrypting the confidential data using a symmetric algorithm such as AES.
3. Protecting the symmetric key using an appropriate public-key mechanism or key-establishment protocol.
4. Sending the encrypted data and protected key to the recipient.
5. Recovering the symmetric key and decrypting the data.

Hybrid encryption combines the efficiency of symmetric encryption with the key-management advantages of public-key cryptography.

Encryption and Privacy in Blockchain

Encryption is particularly important in permissioned blockchains and blockchain applications that handle sensitive information.

For example, in a blockchain-based healthcare system, medical records can be encrypted and stored off-chain, while the blockchain stores a record reference, integrity hash and access-control information. Authorized users can decrypt the records using appropriate keys.

This approach can help protect confidentiality while preserving evidence of data integrity. However, metadata and transaction details may still reveal information unless additional privacy protections are implemented.

Encryption Compared with Hashing and Digital Signatures

Feature	Encryption	Hashing	Digital Signature
Main purpose	Confidentiality	Integrity checking	Authenticity and authorization
Reversible	Yes, with the	No, in normal use	Not a data-recovery

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	appropriate key		mechanism
Key requirement	Depends on the scheme	Usually no secret key for ordinary cryptographic hashing	Private key for signing; public key for verification
Blockchain use	Protecting confidential data	Linking blocks and checking data integrity	Authorizing transactions

Challenges of Encryption in Blockchain

- Key Management: Losing a decryption key may make encrypted information unrecoverable.
- Permanent Storage: Encrypted data placed on a blockchain may remain available for a long time, creating risks if the encryption later becomes vulnerable.
- Performance: Encryption and privacy mechanisms may introduce computational overhead.
- Access Control: Applications must ensure that only authorized users obtain decryption keys.
- Quantum Threats: Some widely used public-key cryptographic mechanisms may become vulnerable to sufficiently powerful quantum computers.

Applications

Encryption schemes can support blockchain-based healthcare records, confidential business transactions, secure supply-chain information, digital identity systems and protected document sharing.

Conclusion: AES, RSA, ECC-based key establishment and hybrid encryption play different roles in protecting confidential information. Selecting an appropriate approach requires consideration of security, performance, key management and the privacy needs of the blockchain application.

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